

JAWAHAR . R

B.Tech Student · Artificial Intelligence & Data Science · Aspiring Data Engineer

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S U M M A R Y

Driven B.Tech student in Artificial Intelligence & Data Science at Dhanalakshmi Srinivasan Engineering College (Autonomous), Perambalur. Hands-on experience building end-to-end data pipelines and real-time ETL systems from client-side weather dashboards to FastAPI-backed ingestion pipelines and NLP-powered assistants for real-world social impact. Certified across Data Engineering, GenAI, and Cloud Analytics by AICTE, GUVI/HCL, NPTEL, and Google. Documents pipeline decisions for reproducibility and evaluates systems against clear metrics. Seeking a Data Engineering Internship where I can apply Python, SQL, ETL, and cloud technologies to build scalable data systems while contributing to real-world products.

E X P E R I E N C E

Aspiring Data Engineer · Independent Builder

2024 – Present

Self-Directed Projects & Hackathons | AI-ML Virtual Intern (AICTE + EduSkills + Google)

- ❖ Built 5+ real-world data engineering & AI projects covering ETL pipeline design, REST API development, real-time data ingestion, and NLP-powered systems.
- ❖ Completed 10-week AI-ML Virtual Internship (AICTE + EduSkills + Google, Oct–Dec 2025); presented Live City Pulse data pipeline at the national AI For Bharat Hackathon and college project expo.

P R O J E C T S

1. FlowMatrix — a Production-Grade End-to-End Real-Time Data Engineering Pipeline -Kafka · PySpark · dbt · Airflow · Great Expectations · Streamlit · PostgreSQL. flowmarics.streamlit.app/.streamlit.app

- Designed and deployed a production-grade, full-stack e-commerce data pipeline modeled on the Brazilian Olist dataset as a Rakuten Ichiba proxy, using Apache Kafka, PySpark, dbt, Apache Airflow, and Great Expectations for ingestion, transformation, and data quality validation.
- Built an interactive Streamlit front-end backed by PostgreSQL (Neon Serverless), migrating the database from a local Docker instance via `pg_dump/psql` and configuring SSL for non-local connections.
- Resolved production deployment issues (dependency conflicts, database connection handling) to ship a live, publicly accessible pipeline central to Rakuten Japan internship preparation.

2. Weather ETL Dashboard-JavaScript · HTML/CSS · OpenWeatherMap API/github.com/j26219096-prog/weather-etl-dashboard

- Engineered a fully client-side ETL pipeline extracting live weather & AQI data from OpenWeatherMap API across 5 global cities; implements Extract → Transform → Load entirely in vanilla JavaScript.
- Transforms raw JSON into normalised metrics and streams data into a real-time animated dark dashboard; zero-backend, zero-dependency, deployed on GitHub Pages.
- Secure API key handling via browser localStorage — key never exposed in source code; fully responsive across desktop, tablet, and mobile.

3. Real-Time 3D Asset Data Pipeline & Serving API-Python · FastAPI · Uvicorn · Three.js · NLP · Render.com

github.com/j26219096-prog/Real-Time-3D-Asset-Data-Pipeline-Serving-API

- Built a full ETL pipeline with FastAPI: extracts binary .glb 3D assets via HTTP streaming, transforms text prompts using NLP keyword-matching into structured metadata queries, loads into a thread-safe local cache.
- Scalable REST API (Render.com) serves cached assets to a Three.js WebGL frontend; implemented OS-level thread concurrency locks reducing repeat-load latency to near-zero.
- Showcases core Data Engineering principles — ingestion, transformation, caching, and API serving — applied to 3D asset delivery.

4. Kisan-Sahayak — AI For Bharat Hackathon-Python · NLP · Translation API · Market Data API

github.com/j26219096-prog/Kisan-Sahayak

- AI-powered agricultural assistant for rural Indian farmers (Team Kisan-Tech): delivers real-time crop market prices and expert farming advice in Tamil & Hindi using an NLP translation pipeline.
- Selected for national-level AI For Bharat Hackathon — bridges India's digital divide in agri-tech through accessible, local-language AI.

C E R T I F I C A T I O N S

❖ Computer Architecture & Organization Oct 2025	NPTEL / IIT Kharagpur	Jul–
❖ AI-ML Virtual Internship Oct–Dec 2025	AICTE + EduSkills + Google for Developers	
❖ GenAI Powered Data Analytics Job Sim. 22, 2025	Tata / Forage	Dec
❖ Introduction to Data Engineering 7, 2026	IBM / Coursera	May
❖ Python for Data Science, AI & Development 2026	IBM / Coursera	Jul 6,
❖ Python Project for Data Engineering 22, 2026	IBM / Coursera	Jul
❖ Google Cloud Data Analytics Certificate 2026	Google Cloud	Jul

S K I L L S

- **Languages** -- Python · SQL · JavaScript · HTML / CSS
- **Frameworks** -- FastAPI · Uvicorn · Three.js · REST APIs
- **Data Engineering** -- ETL Pipelines · Data Ingestion · Data Caching · API Design · Real-Time Processing
- **AI / ML** -- NLP · Computer Vision · Machine Learning · Generative AI
- **Tools** -- Git/GitHub · GitHub Pages · Render.com · VS Code · Google Analytics · Excel / Google Sheets

E D U C A T I O N / L A N G U A G E S / K E Y A C H I E V E M E N T S

- **B.Tech — Artificial Intelligence & Data Science** **2024 – 2028**
Dhanalakshmi Srinivasan Engineering College (Autonomous), Perambalur, Tamil Nadu

LANGUAGES:

- **Tamil — Native**
- **English — Advanced**
- **Japanese — Beginner (Currently preparing for JLPT N5)**

KEY ACHIEVEMENTS:

- **Built 6+ production-ready data engineering and AI projects.**
- **National-level AI for Bharat Hackathon participant.**
- **Google Cloud Data Analytics Certified.**
- **Publicly deployed ETL applications.**